WeGo Public Transit Analysis

Nova

Shannon, Josh, Sivaraja

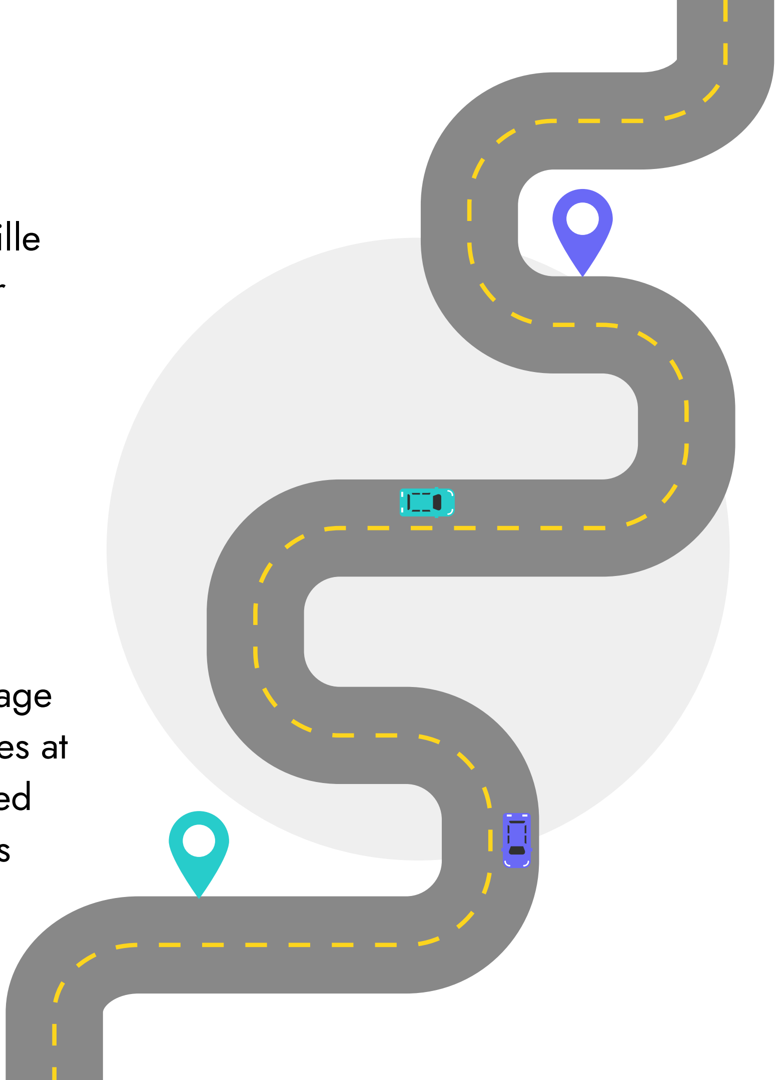
Project Overview

The WeGo Public Transit system serves the Greater Nashville and Davidson County areas. The data was provided by our instructor.

The data consisted of the following:

- Trips on Route 50, Charlotte Pike
- Directions to and from Downtown Nashville
- Years 2023 -2025

Transit Signal Priority (TSP) is a technology that helps manage traffic flow more efficiently. For buses, it reduces wait times at traffic signals by holding green lights longer, shortening red lights or in some cases allowing busses to bypass. TSP was active during specific times and sections of this route.



Main Research Question

Does Transit Signal Priority (TSP) reduce the probability that a bus leaves a stop late?

Method and Technology

R

- TidyVerse
- Lubridate
- ggplot2
- Dplyr

Logistic Regression

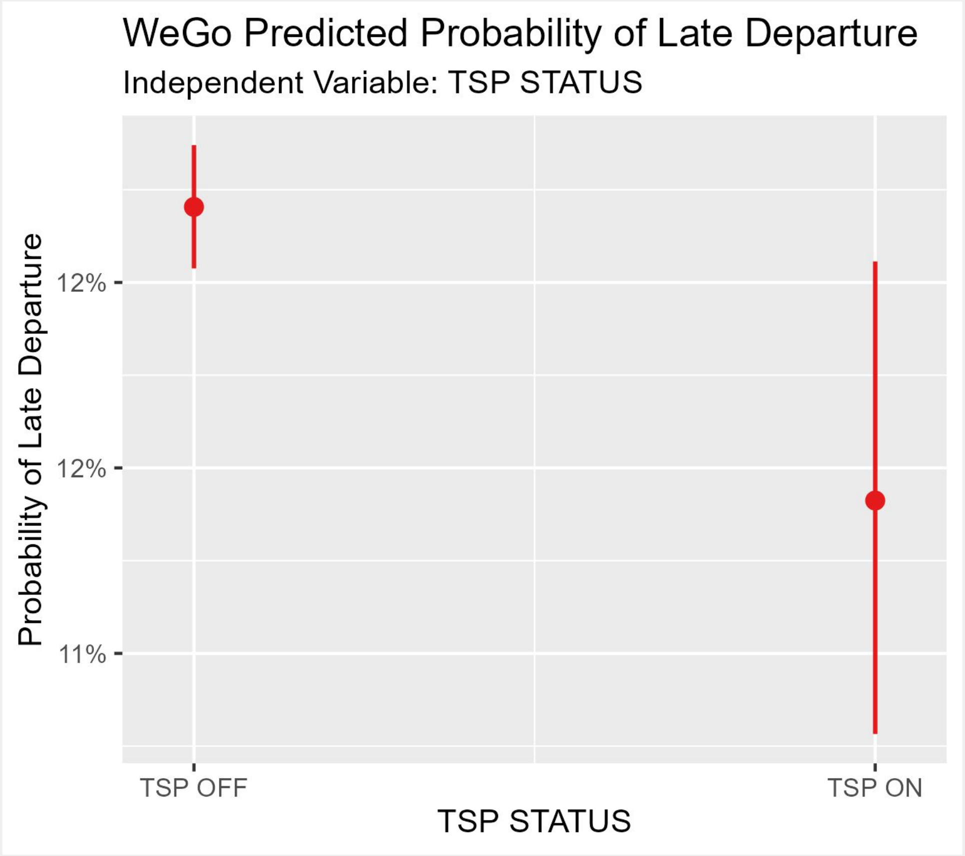
```
glm(
    ADJUSTED_LATE_COUNT ~ TSP_STATUS,
    data=wego,
    family="binomial"
)
```

WeGo Data

TSP Status

White Bridge ⟷ MCC

Simple Logistic Regression Model



Late Departure ~ TSP Status

Qualitative independent variable

p-value = **0.02**

The p-value is < 0.05, so we can conclude that the TSP Status is **significant**.

Variables in Approach

Time of Day
Compared travel times that can be expected based on the hour of the day.

Day of the Week
Checked each day of the week to compare weekdays and weekends.

The Year 2025
Filtered down our date to only include the year 2025.

Overload
Selected routes that were adding a stop to their original route or not, mostly due to maintenance.

Canceled Stop
Compare routes that had a stop canceled on their original route.

Time of Day

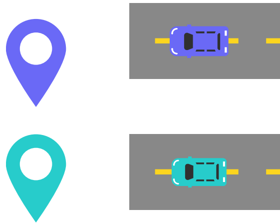
Each part of the day was divided up by the hour.

- hour >= 4 & hour <= 5 ~ "early_morning",
- hour >= 6 & hour <= 8 ~ "morning_peak",
- hour >= 9 & hour <= 14 ~ "midday",
- hour >= 15 & hour <= 17 ~ "pm_peak",
- hour >= 18 & hour <= 20 ~ "evening",
- hour >= 21 | hour <= 1 ~ "late_night",

Then in the model we used, we added the time of day as a variable to our logistic regression model.

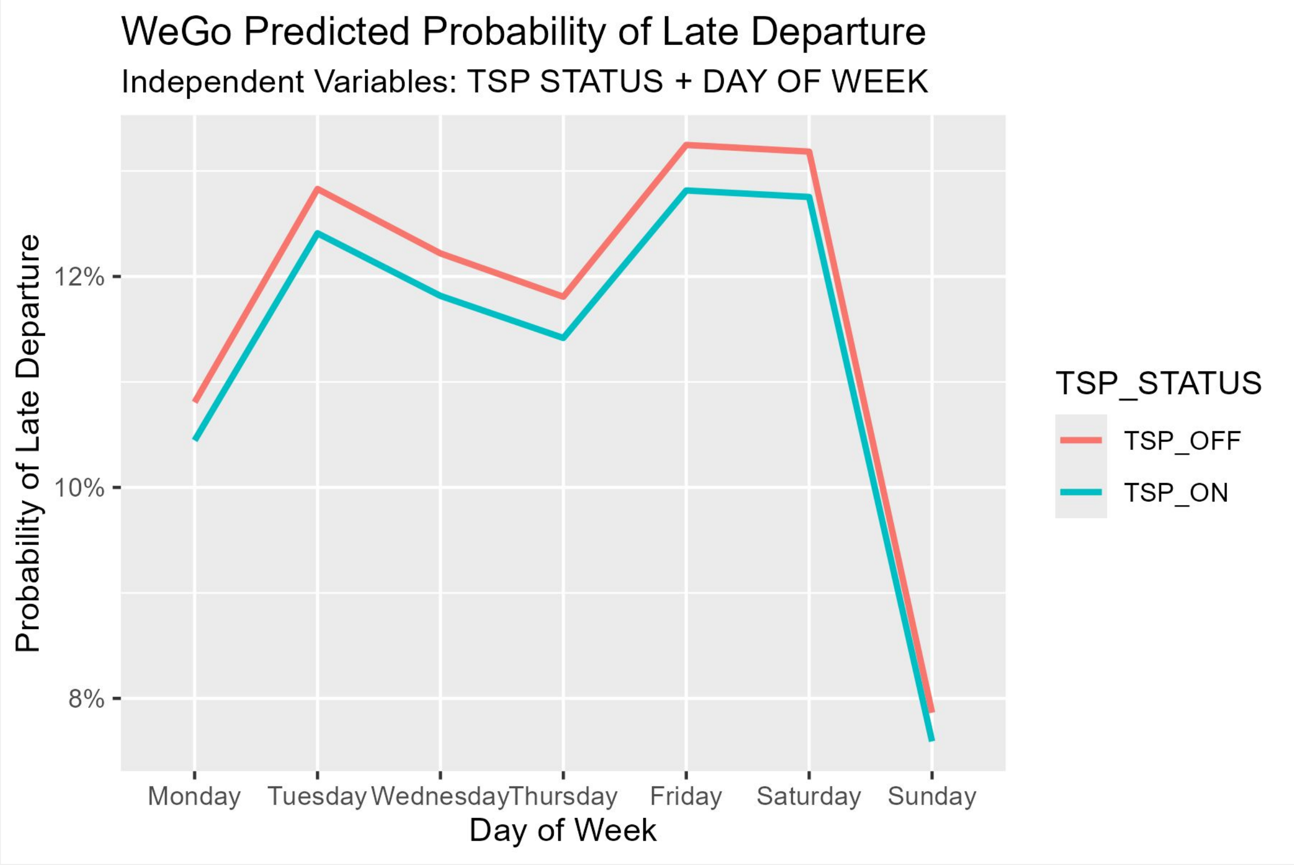
```
wego_time_of_day_lr <- glm(ADJUSTED_LATE_COUNT ~ TSP_STATUS + TIME_OF_DAY,
                           data=wego, family="binomial")
```

Predicted Probability of Late Bus Departure by Time of Day



This figure shows that predicted lateness is slightly lower when TSP is active, particularly during peak travel periods.

Late Departure ~ TSP Status



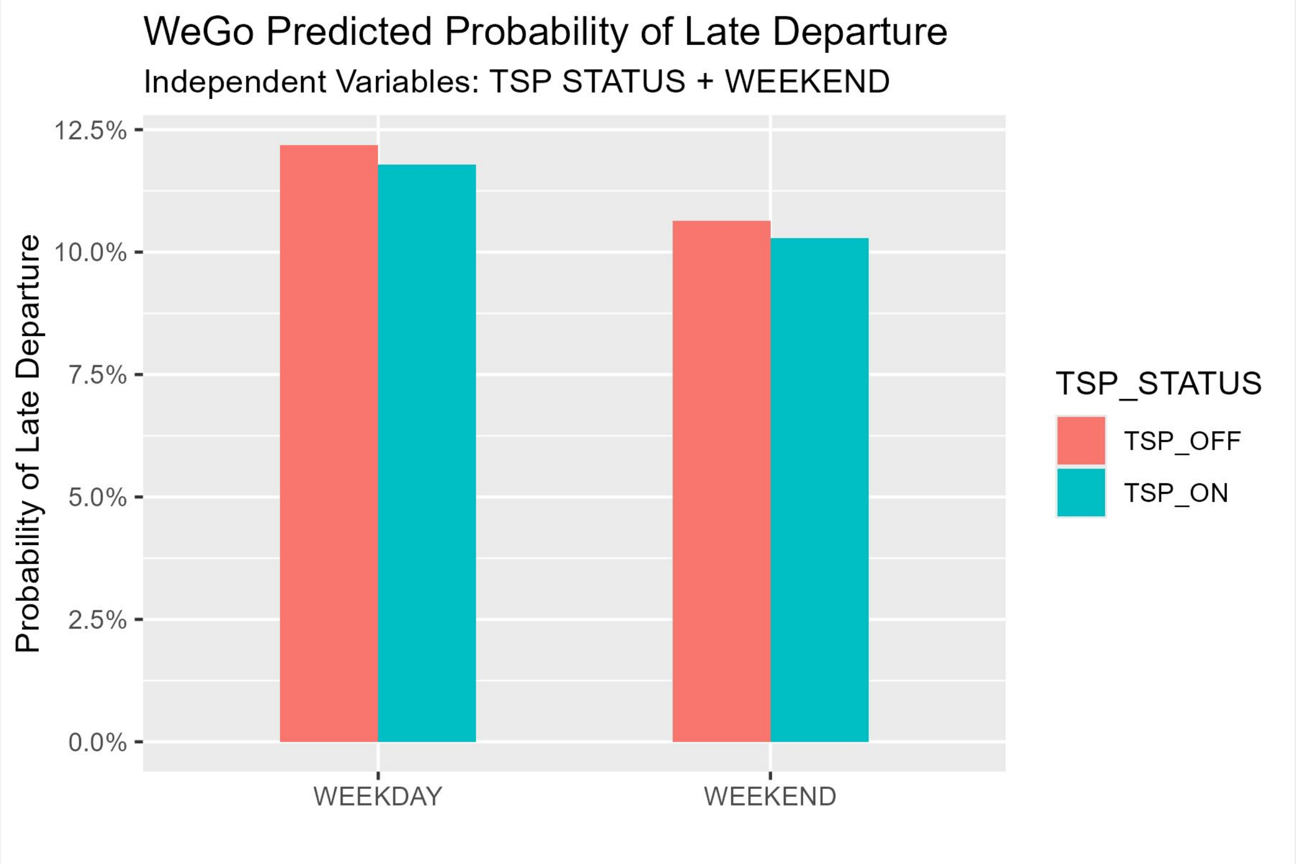
+ Day of the Week

p-values

Monday	< 2e-16
Tuesday	0.00628
Wednesday	1.11e-11
Thursday	< 2e-16
Friday	0.02130
~~Saturday~~	~~0.70547~~
Sunday	< 2e-16

The p-values are < 0.05, so we can conclude that these are **significant**.

Late Departure ~ TSP Status



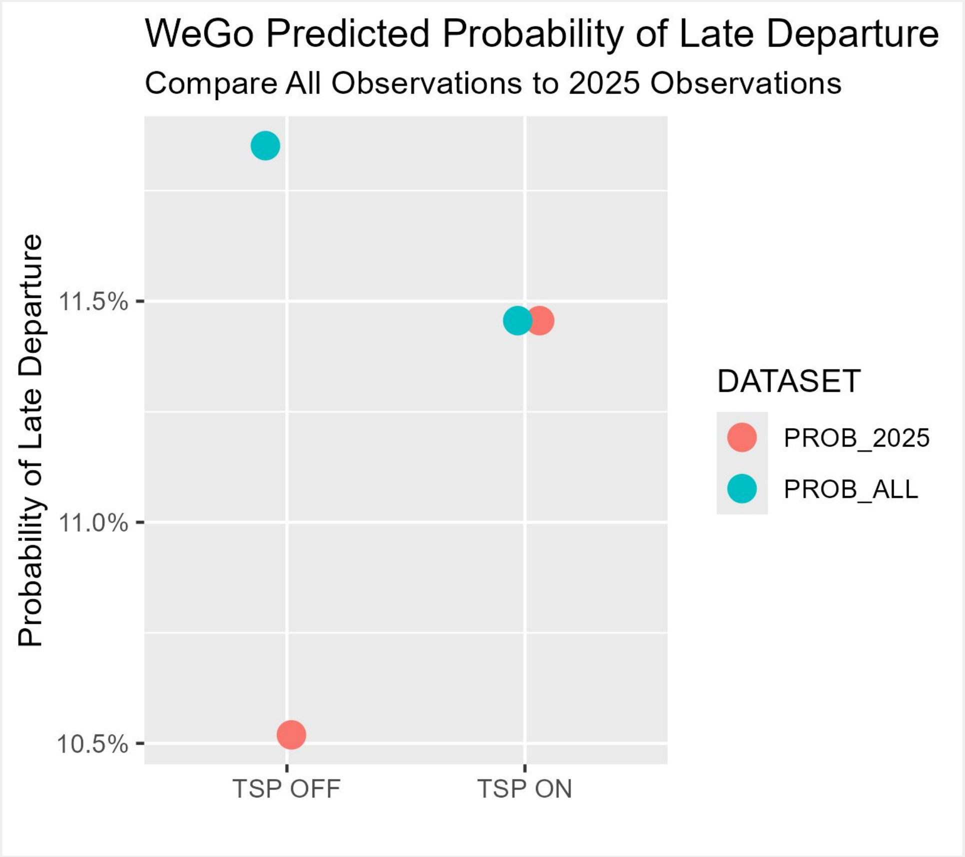
+ Weekend

p-values

Weekday	0.0198
Weekend	< 2e-16

The p-values are < 0.05, so we can conclude that these are **significant**.

Late Departure ~ TSP Status (2025)

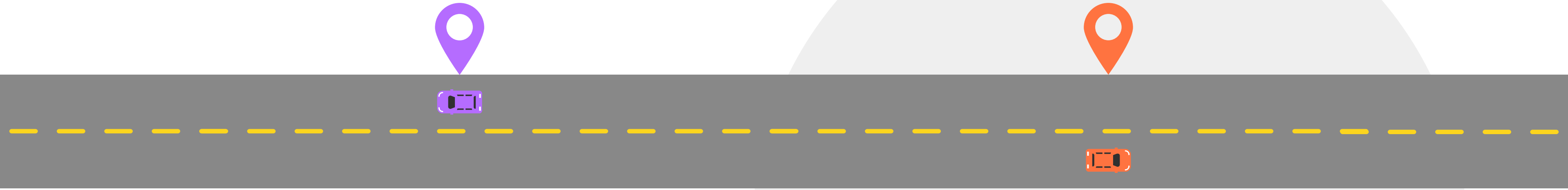


Late Departure ~ TSP Status

p-value = 5.35e-06

The p-value is < 0.05, so we can conclude that the TSP Status is **significant**.

Overload



In the Data

Signifies that the record is from a trip that was added by the dispatcher and was not part of the original schedule for the day. Usually, these are created when one vehicle breaks down and another is covering the same service.

Logistic Model

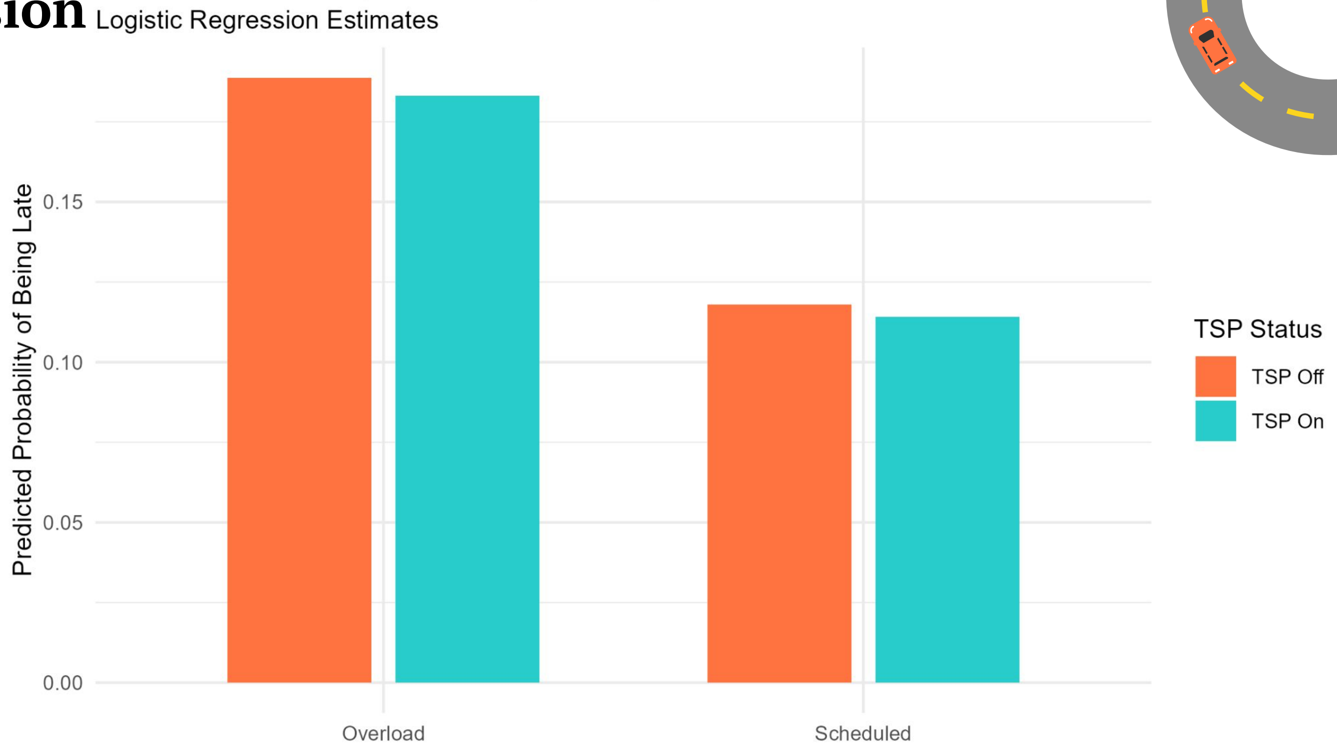
```
wego_overload_id_lr <-

glm(ADJUSTED_LATE_COUNT ~ TSP_STATUS + OVERLOAD_ID, data=wego, family="binomial")
```

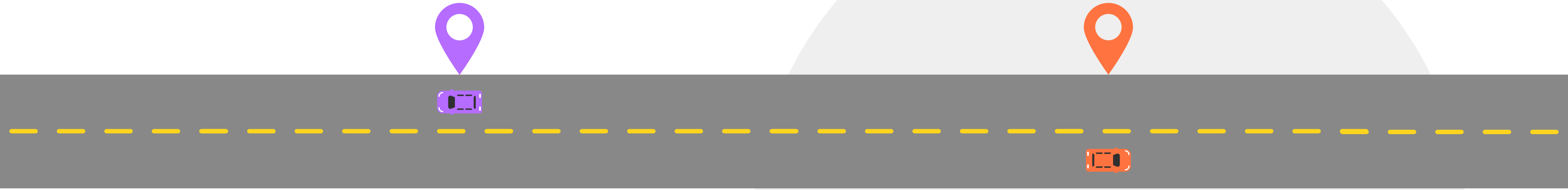
Predicted Probability of Late Departure by Overload Status

Logistic Regression

Predicted probabilities show that overload trips are substantially more likely to be late, but TSP reduces lateness for both scheduled and overload service.



Cancelled Stop



In the Data

Flags whether a crossing was canceled or waived.

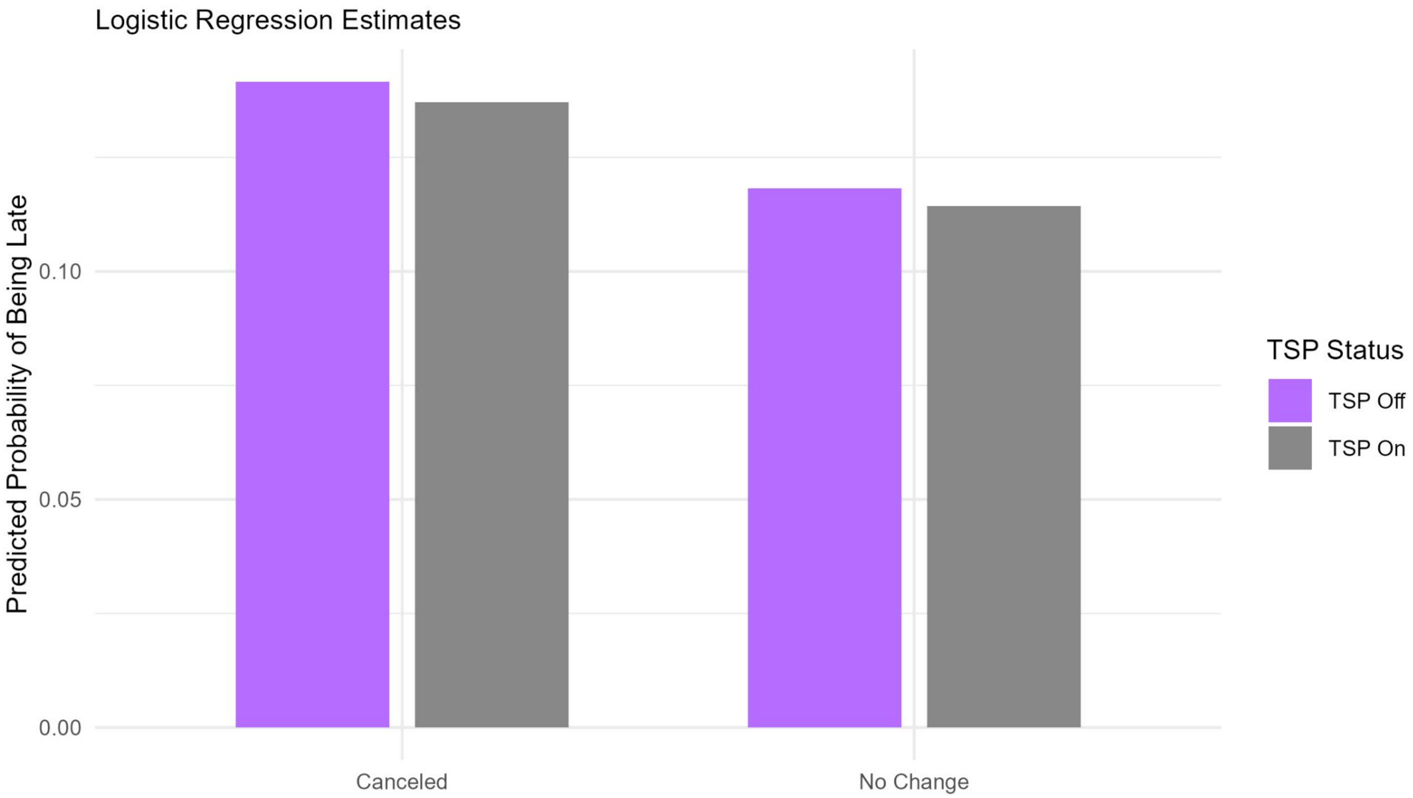
Logistic Regression

```
wego_stop_cancelled_lr <-

glm(ADJUSTED_LATE_COUNT ~ TSP_STATUS +
STOP_CANCELLED, data=wego, family="binomial")
```

Predicted Probability of Late Departure by Stop Cancelation Status

Logistic Regression Estimates



In summary:

Cancelled stops are associated with a higher predicted probability of lateness, but TSP reduces lateness regardless of cancellation status. This was statistically significant.

Conclusion

Does Transit Signal Priority (TSP) reduce the probability that a bus leaves a stop late?

Yes

Overall, yes. We noticed there were more “on time” busses when TSP was active.

Why TSP?

TSP makes traffic for busses more efficient.

Time

Overall difference approx. 0.193%

Largest difference Weekday approx. 1.544%

Further Study

We would want to know the return on investment for TSP.

We want to also see how often people use the busses for transportation in the city.

Route Direction